$t_n \in \{-1, 1\}$
binary classification

Hard-Margin SVM

$$\arg\min_{\{w, w_0\}} \frac{1}{2}\|w\|^2 \quad \text{subject to} \quad t_n \cdot y(x_n) \geqslant 1, \ \forall x_n$$

$$\Leftrightarrow \underbrace{t_n \cdot y(x_n) - 1}\ \geqslant 0$$

$$y(x_n) = w^T \cdot \phi(x_n) + w_0$$

$N \equiv$ # training samples

Using the Lagrange Optimization:

$\vec{a} = (a_1, a_2, \dots, a_N)$

$$\boxed{\mathcal{L}(w, w_0, \vec{a}) = \frac{1}{2}\|w\|^2 - \sum_{n=1}^{N} a_n \cdot (t_n \cdot y(x_n) - 1)}$$

Minimizing this function!

$$y(x_n) = w^T \phi(x_n) + w_0$$

with KKT conditions
$$\begin{cases} t_n \cdot y(x_n) - 1 \geqslant 0 \\ a_n \geqslant 0 \\ a_n (t_n \cdot y(x_n) - 1) = 0 \end{cases}, \forall n$$

1

$t_n \cdot y(x_n) > 1$

$a_n = 0$

$\downarrow$

x_n is NOT a S.V.

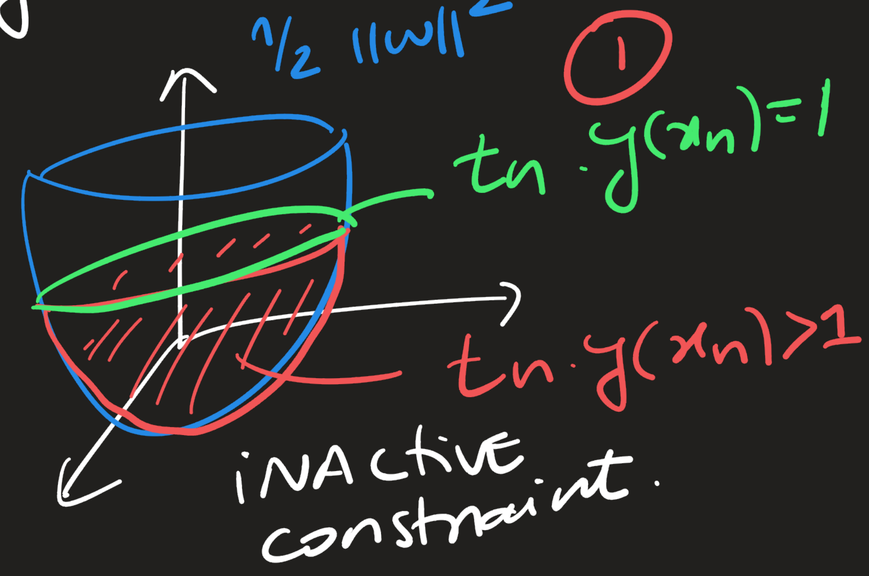
2

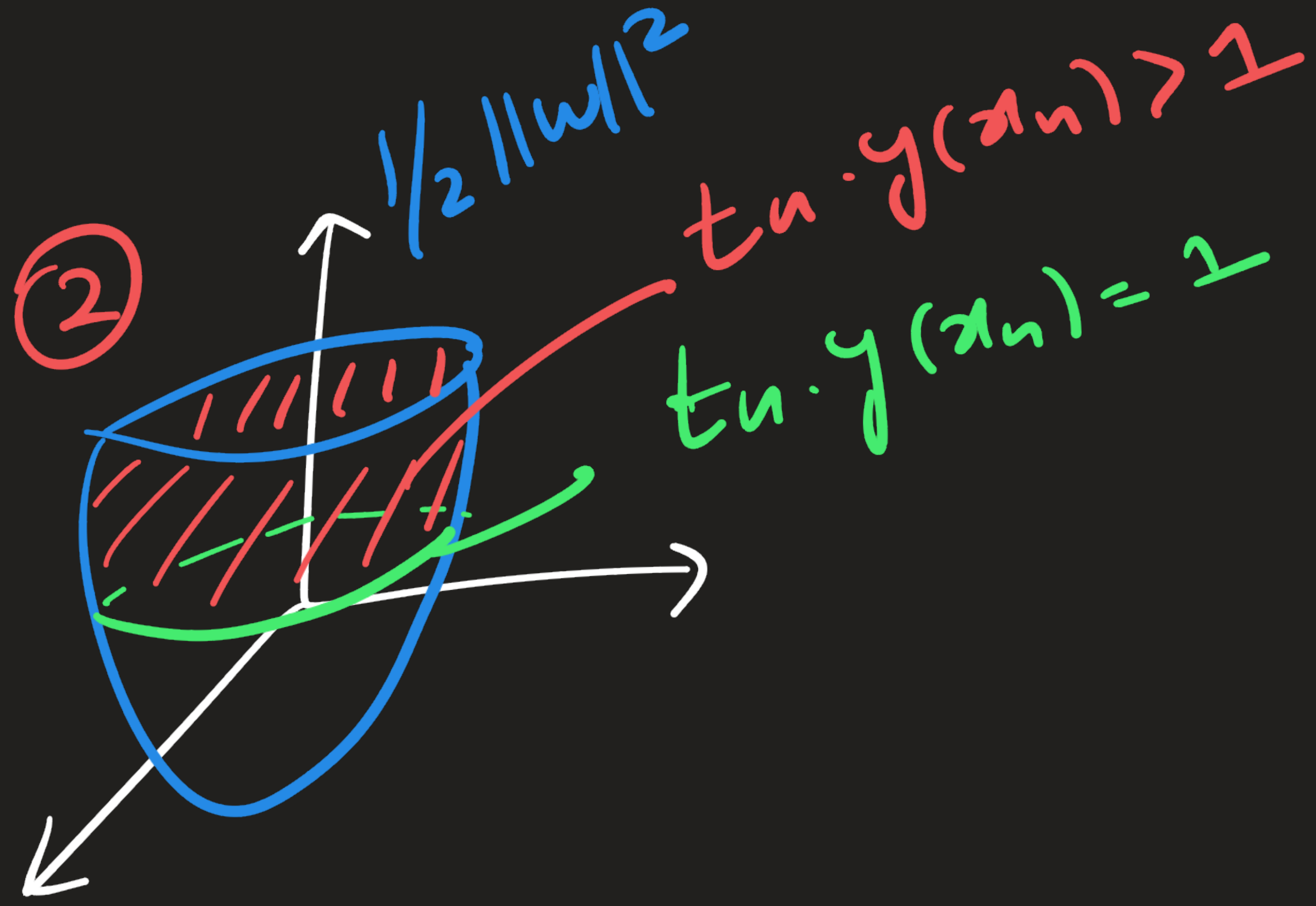
$t_n \cdot y(x_n) = 1$

$a_n > 0$

$\downarrow$

x_n must be a support vector (S.V.)





Active constraint

optimizing for $\mathcal{L}$ w.r.t. w and w_0:

$$\begin{cases} \dfrac{\partial \mathcal{L}}{\partial w} = 0 \\ \dfrac{\partial \mathcal{L}}{\partial w_0} = 0 \end{cases} \Leftrightarrow \begin{cases} \boxed{w = \sum_{n=1}^{N} a_n . t_n . \phi(x_n)} \\ \sum_{n=1}^{N} a_n . t_n = 0 \end{cases}$$

we can substitute this solution for w in the mapper function:

$$\boxed{y(x)} = w^T \phi(x) + w_0$$

$$= \left[\sum_{n=1}^{N} a_n . t_n . \overbrace{\phi^T(x_n)] \phi(x)}^{K(x_n, x) \equiv \text{KERNEL function}} + w_0$$

test sample

training sample

$$= \Big(a_1 . t_1 . \phi^T(x_1) . \phi(x)$$
$$+ a_2 . t_2 . \phi^T(x_2) . \phi(x)$$
$$\vdots$$
$$+ a_n . t_n . \phi^T(x_n) . \phi(x) \Big) + w_0$$

$\phi(x)$ is an appropriate transformation function if the resulting kernel function (matrix) is semi-positive definite matrix.

For a given training set $\{x_i\}_{i=1}^{N}$:

The kernel matrix looks like:

$$K = \begin{bmatrix} \phi(x_1)^T \phi(x_1) & \phi(x_1)^T \phi(x_2) & \cdots & \phi^T(x_1) \cdot \phi(x_N) \\ \phi^T(x_2) \cdot \phi(x_2) & \phi(x_2)^T \phi(x_2) & \cdots & \phi^T(x_2) \cdot \phi^T(x_N) \\ \vdots & & & \\ \phi(x_N) \cdot \phi^T(x_1) & \phi(x_N)^T \cdot \phi(x_2) & & \phi^T(x_N) \cdot \phi(x_N) \end{bmatrix}$$

NXN

1. symmetric
2. semi-positive definite
   a) for a vector x: $x^T K . x \geq 0, \forall \vec{x} \setminus \{\vec{0}\}$
   b) All eigenvalues of K are ≥ 0.

In some cases, $\phi(\cdot)$ is infinite-dimensional, thus it cannot be computed or stored.

But if we select a proper function, the $K(x, x_n) = \phi(x)^T \cdot \phi(x_n)$ is a finite quantity that measures similarity in the space spanned by $\phi(x)$ without us having to compute it.

$\Longrightarrow$ Kernel Trick.

Transformation fct $\phi : \mathbb{R}^2 \longrightarrow \mathbb{R}^3$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \longmapsto \begin{bmatrix} x_1^2 \\ \sqrt{2} \cdot x_1 \cdot x_2 \\ x_2^2 \end{bmatrix}$$

Kernel fct : $K : \mathbb{R}^2 \times \mathbb{R}^2 \longrightarrow \mathbb{R}$

$$(x, y) \longmapsto \phi^T(x) \cdot \phi(y)$$

$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $\vec{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$

$$\boxed{K(\vec{x}, \vec{y})} = \phi^T(x) \cdot \phi(y)$$

$$= [x_1^2, \sqrt{2} \cdot x_1 \cdot x_2, x_2^2] \begin{bmatrix} y_1^2 \\ \sqrt{2} \cdot y_1 \cdot y_2 \\ y_2^2 \end{bmatrix}$$

$$= x_1^2 \cdot y_1^2 + 2 \cdot x_1 \cdot x_2 \cdot y_1 \cdot y_2 + x_2^2 y_2^2$$

$$= (x_1 \cdot y_1 + x_2 \cdot y_2)^2$$

$$= \left([x_1, x_2] \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \right)^2$$

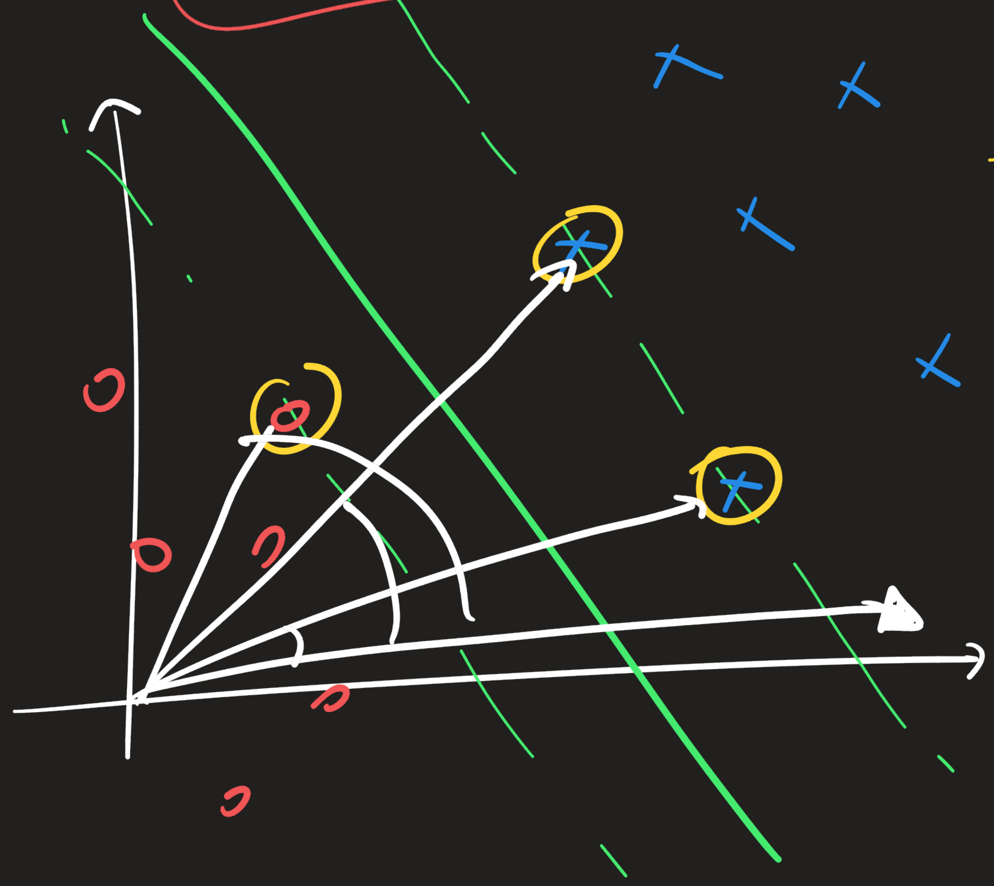
$$\boxed{= (x^T \cdot y)^2}$$

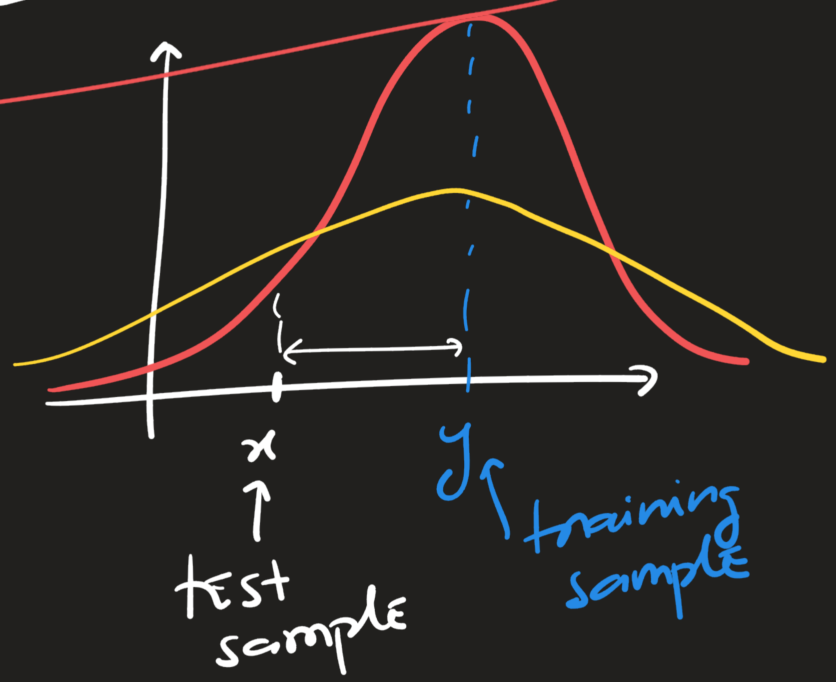
Another popular kernel is the Radial Basis Function (RBF)

kernel:

$$\boxed{K_{RBF}(x,y) = e^{-\frac{1}{2\sigma^2}\cdot\|x-y\|^2}}$$

$\Longrightarrow$ Associated ϕ_{RBF} is infinite-dimensional





$$y(x) = \sum_{n=1}^{N} a_n \cdot t_n K(x_n, x) + w_0 \in \mathbb{R}$$

$$\begin{cases} y(x) > 0 \Rightarrow x \in C_1 \\ y(x) < 0 \Rightarrow x \in C_{-1} \end{cases}$$